Ch 10.1: Neural Nets

Lecture 29 - CMSE 381

Prof. Elizabeth Munch

Michigan State University
::
Dept of Computational Mathematics, Science & Engineering

Mon, Nov 18, 2024

Announcements

Last time:

- SVM

This lecture:

- Feed Forward Neural Nets

Announcements:

- Homework #9 due Sunday, Nov 24
- No class wednesday Nov 27. Virtual office hours by request.

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #6 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #7 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #8 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

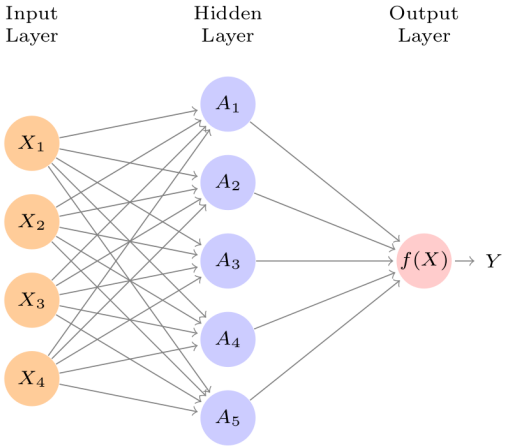
Section 1

Neural Nets

The idea

- Draw input of vector p variables $X = (X_1, \cdots, X_p)$
- Draw black box nonlinear function $f(X)$
- Spit out response prediction Y
- We've had nonlinear stuff before like trees, boosting, GANs
- Difference here is the structure

Feed Forward Neural Network: The cartoon

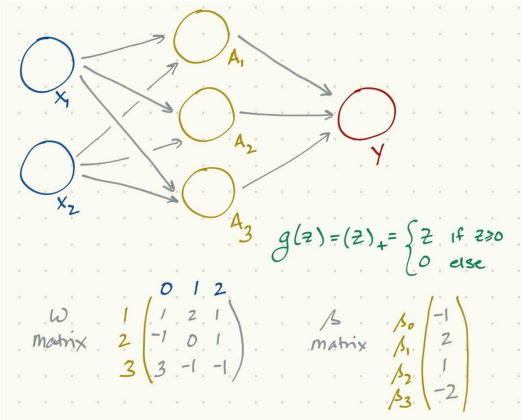


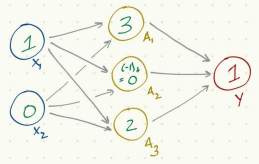
- Input layer is starting data
- Each arrow means taking combo of those values with weights to get next value
- Here there is one hidden layer then the output
- Get to pick how many hidden units K, here we have $K = 5$

A very simple example

Computing A_k for $(1, 0)$

$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$





$k=1$

$\omega_{10} + \omega_{11} X_1 + \omega_{12} X_2$

$1 + 2(1) + 1(0) = 3$

$\beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3$

$k=2$

$\omega_{20} + \omega_{21} X_1 + \omega_{22} X_2$

$-1 + 0(1) + 0(0) = -1$

$= -1 + (2)(3) + (1)(0) + (-2)\,2$

$= -1 + 6 + 0 - 4$

$= 1$

$k=3$

$\omega_{30} + \omega_{31} X_1 + \omega_{32} X_2$

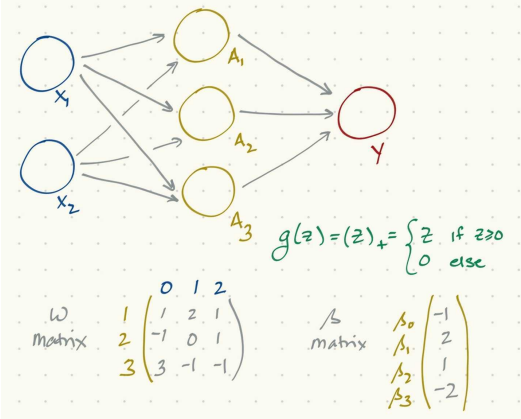
$3 + (-1)(1) + (-1)(0) = 2$

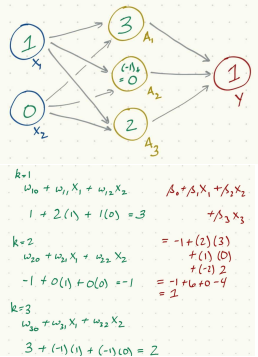
- A_k called activations
- Pick an activation function $g(z)$

A very simple example

Computing Y for $(1, 0)$

$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

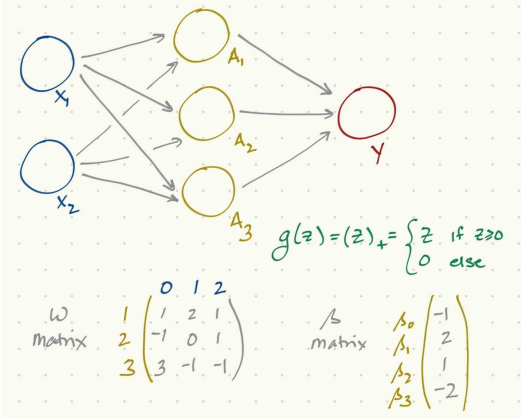


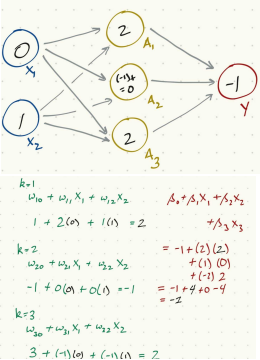


A very simple example

Computing Y for $(0, 1)$

$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j), \qquad f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$





A different example

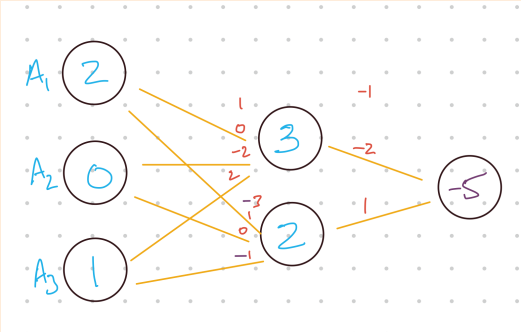
- Draw the diagram for a neural net with input data points with $p = 3$ (i.e., (X_1, X_2, X_3)) and two units in the hidden layer.
- Using the ω and β matrices, what is the output predicted Y for the point $(2, 0, 1)$?

$$\omega = \begin{pmatrix} 1 & 0 & -2 & 2 \\ -3 & 1 & 0 & -1 \end{pmatrix} \qquad \beta = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$$

- Use the activation function

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$

Extra space



Oops, those three shouldn't be labeled as A's, it should be X's. Also check the right, I did the relu computation wrong so some error propagated!

$$A_k = g\left(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j\right)$$

$$A_1 = g(w_{10} + w_{11} X_1 + w_{12} X_2 + w_{13} X_3)$$

$$A_1 = g(1 + 0 \cdot 2 + -2 \cdot 0 + 2 \cdot 1)$$

$$g(1 + 2) = g(3) = 3$$

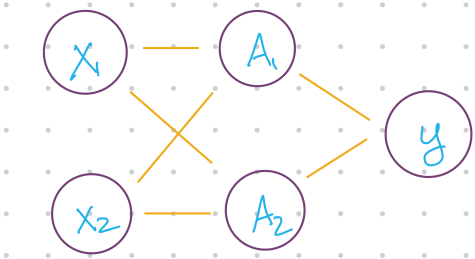
$$A_2 = g(-3 + 1 \cdot 2 + 0 \cdot 0 - 1 \cdot 1)$$

$$= g(-3 + 2 + 0 - 1) = g(-2) = 2$$

$$\beta_0 + \beta_1 A_1 + \beta_2 A_2$$

$$-1 - 2 \cdot 3 + 1 \cdot 2 = -7 + 2 = -5$$

What if there's no activation function?



$$\omega = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \qquad \beta = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$$

Calculate with no activation function to show that the result is just some other linear function

$A_1 = 1 + 2X_1 + 3X_2$
$A_2 = 4 + 5X_1 + 6X_2$

$1 + 2(A_1) + 1(A_1) =$

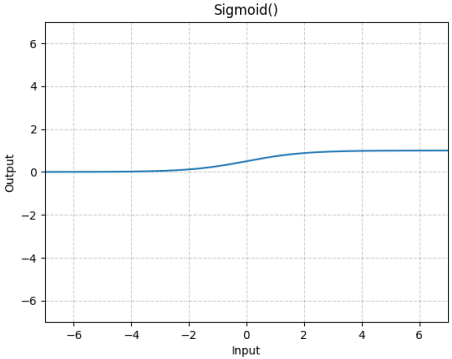
$= 1 + 2(1 + 2X_1 + 3X_2) + (4 + 5X_1 + 6X_2)$

$= 7 + 9X_1 + 12X_2$

Choices for activation function

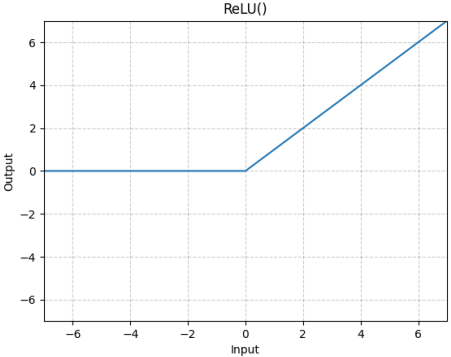
Sigmoid:

$$g(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

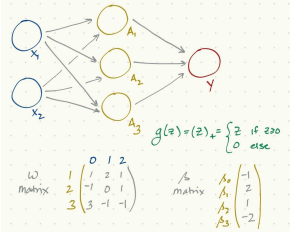


ReLU: Rectified linear unit

$$g(z) = (z)_+ = \begin{cases} 0 & \text{if } z < 0 \\ z & \text{else.} \end{cases}$$



Matrix version: First layer

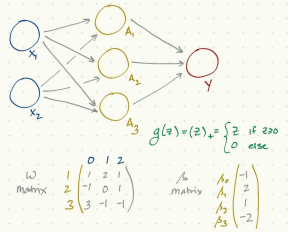


$$A_k = h_k(X) = g(w_{k0} + \textstyle\sum_{j=1}^{p} w_{kj} X_j),$$

$$A = g(\mathbf{W} \cdot \mathbf{X}) \qquad \mathbf{X}^T = (1\ X_1\ X_2\ \cdots\ X_p)$$

Caclulate out the matrix multiplication using the matrices at left to show that the equations are the same.

Matrix version: Output



$$f(X) = \beta_0 + \sum_{k=1}^{K} \beta_k A_k$$

$$Y = \beta \cdot \mathbf{A} \qquad \mathbf{A}^T = (1 \; A_1 \; A_2 \; \cdots \; A_K)$$

Now what?

Choose parameters by minimizing RSS, $\sum_{i=1}^{n}(y_i - f(x_i))^2$

Chosen in advance:

- Number of layers (more on that next lecture)
- Number of hidden units
- Activation function $g(z)$

Tuned by the model:

- ω
- β

Coding

Next time

Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #6 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #7 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #8 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due